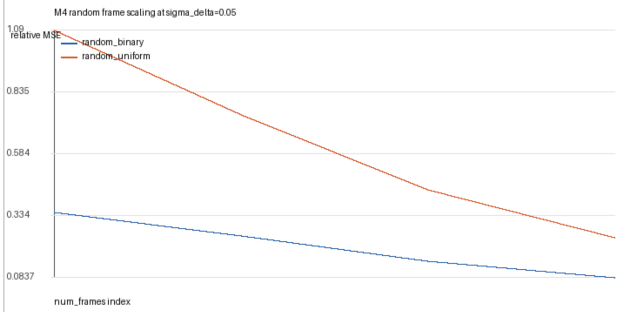
Figure 4. Residual gain errors and averaging laws

Residual gain error scales quadratically, while random-frame reconstructions average residual error down with frame count.

Status: strong quantitative support

B Random-frame curve

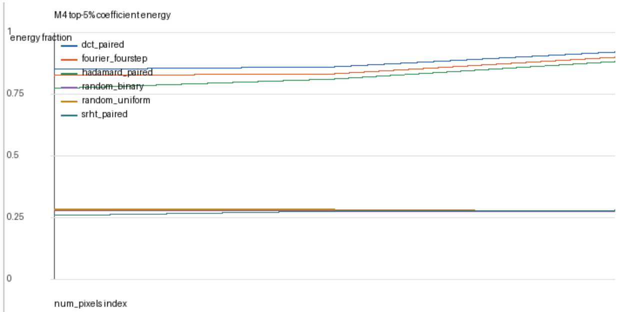


A Residual gain scaling

M4 residual gain error scaling

basis	sigma_delta_exponent	sigma_delta_exponent_ci_low	sigma_delta_exponent_ci_high	num_frames_exponent	r2
dct_paired	2.0017	1.9892	2.0117	0.78811	0.99996
fourier_fourstep	2.0014	1.9914	2.0126	0.87949	0.99995
hadamard_paired	2.0012	1.99	2.0124	0.78596	0.99996
random_binary	2.0018	1.9901	2.0144	0.74883	0.99992
random_uniform	2.0028	1.9964	2.0117	0.81144	0.99997
srht_paired	2.0013	1.9899	2.012	0.78639	0.99996

D Energy concentration



C Frame-count fit

M4 random frame scaling fits

basis	sigma_delta_exponent	num_frames_exponent	num_frames_exponent_ci_low	num_frames_exponent_ci_high	r2
random_binary	1.9982	-0.70985	-0.77224	-0.65886	0.99872
random_uniform	1.9993	-0.72216	-0.74877	-0.69518	0.99964

The main fitted laws supporting H2/H4: residual gain errors have near-quadratic scaling; random-frame correlation reconstructions improve with frame count; deterministic bases concentrate coefficient energy.